SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

(192411091)
I POOJASREE B *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Poojasree B

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ARTIFICIAL INTELLIGENCE

CO1_AT2

POOJASREE B
192411091
CSA1710

1. **Backtracking Search - Delivery Truck Scheduling**

Given

Trucks: T_1, T_2, T_3

Time windows

A = 9-11 AM

B = 11AM - 1 PM

C = 1 - 3 PM

Contraints

1. $T_1 \neq C$
2. T_2 must be before T_3
3. Only one truck per time window
4. $T_3 \neq A$
5. Every truck gets exactly one time window

Step 1: Domain

Truck	Possible Time Window
T_1	A, B
T_2	A, B, C
T_3	B, C

Step 2: Backtracking Search

Assign $T_1 = A$

Current Assignment: $T_1 = A$

Remaining:

$T_2 \rightarrow \{B, C\}$

$T_3 \rightarrow \{B, C\}$

Try $T_2 = B$

Current Assignment: $T_1 = A$
$T_2 = B$

Remaing: $T_3 = C$

Constraint Check

$T_1 \neq C$

$T_3 \neq A$

T_2 before T_3 (B before C)

One truck per slot

Search Tree

- Start
  - $T_1 = A$
    - $T_2 = B$
      - $T_3 = C$ ✓
    - $T_2 = C$

      (T_2 not before T_3)
  - $T_1 = B$
    - $T_2 = A$
      - $T_3 = C$ ✓
    - $T_2 = C$ ✗

Final Valid Schedule

Truck	Time
Truck 1	9-11 AM
Truck 2	11AM - 1PM
Truck 3	1-3 PM

2. BFS Search

Assumed 5×5 Grid

	C_1	C_2	C_3	C_4	C_5
R_1	S	→	→	↓	·
R_2	·	x	·	↓	·
R_3	·	·	x	→	·
R_4	·	·	·	·	·
R_5	·	·	·	·	G

where

S = Start (1,1)

G = Goal (5,5)

x = Blocked cell

Cost of every move = 1

Movement allowed : Up, Down, Left, Right

One-way restrictions:

(1,2) → Right only

(1,3) → Right only

(1,4) ↓ Down only

(2,4) ↓ Down only

(3,4) → Right only

Manhattan Distance

$$h(n) = |x_g - x| + |y_g - y|$$

Goal (5,5)

Example:

Start $h(s) = |5-1| + |5-1| = 8$

BFS Search

Step 1

Queue [s]

Expand (1,1)

Visited (1,1)

Add (1,2)
(2,1)

Queue [(1,2),(2,1)]

Step 2

Expand (1,2)

Allowed only Right

Add (1,3)

Queue [(2,1),(1,3)]

Step 3

Expand (2,1)

Add (3,1)

Queue [(1,3),(3,1)]

Step 4

Expand (1,3)

Allowed Right

Add (1,4)

Queue [(3,1),(1,4)]

Step 5.

Expand (3,1)

Add (4,1)

(3,2)

Queue [(1,4),(4,1),(3,2)]

Step 6

Expand (1,4)

Allowed Down

Add (2,4)

Queue [(4,1),(3,2),(2,4)]

Step 7

Expand (4,1)

Add (5,1)

(4,2)

Queue [(3,2),(2,4),(5,1),(4,2)]

Step 8

Expand (3,2)

Right is blocked

No new node

Queue [(2,4),(5,1),(4,2)]

Step 9

Expand (2,4)

Allowed Down

Add (3,4)

Queue [(5,1),(4,2),(3,4)]

Step 10

Expand (3,4)

Allowed Right

Add (3,5)

Queue [(5,1),(4,2),(3,5)]

Step 11

Expand (3,5)

Add (4,5)

Queue [(5,1),(4,2),(4,5)]

Step 12

Expand (4,5)

Add (5,5)

Queue [(5,1),(4,2),(5,5)]

Step 13

Expand (5,5) , Goal Found

Optimal path

S → (1,2) → (1,3) → (1,4) → (2,4) ↓

(3,4)

G ← (4,5) ← (3,5) ← (3,4)

Cost calculation

Each moves costs 1 .

Number of moves: 8

Total cost : 8 × 1 = 8

Manhattan Heuristic Values Along the path

Node	h(n)
(1,1)	8
(1,2)	7

(1,3)	6
(1,4)	5
(2,4)	4
(3,4)	3
(3,5)	2
(4,5)	1
(5,5)	0

Final Answer

Optimal Path: (1,1) → (1,2) → (1,3) → (1,4) → (2,4) → (3,4) → (3,5) → (4,5) → (5,5)

Total cost : 8

Goal Reached : Yes

Search Algorith : BFS

Heuristic Used : Manhattan Distance

$$h(n) = |x_g - x| + |y_g - y|$$